



HRITHIK RAI

Founding AI Agent Engineer | Researcher

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 [Portfolio](#)

 [Linkedin](#)

 [Github](#)

PROFILE INFO

An applied AI researcher and systems engineer with over 7 years of experience bridging advanced machine learning with production-grade engineering across government infrastructure, industrial AI labs, and international startups. My research and development portfolio spans the evolution of artificial intelligence, from early applied machine learning to large-scale agentic systems across India, Germany, and Italy. I specialize in the architecture, deployment, and optimization of highly reliable AI systems, with a core focus on compute efficiency, operational sustainability, and algorithmic rigor.

EXPERIENCE

2026 - Present

nexwise | Munich, Germany

Founding AI Agent Engineer

- Development of agent evaluation suite from defining Behaviour Driven Development, to writing test cases and executing test runs.
- Optimizing tool calling around knowledge graphs
- Retriever tuning and evaluation pipeline setup
- Agent optimization over quality gates

2025 - 2026

Estyl | Milan, Italy

Lead AI Software Engineer

- Built the Estyl Agent Service from the ground up, modular multi agentic architecture, REST & WebSocket APIs, long term memory, event tracking and CI/CD. (p95 latency = 2.8 seconds)
- Architected and deployed end-to-end GCP data pipelines for knowledge extraction, from ingestion (SFTP, CSV, JSON, XML parsing) to BigQuery and Vertex AI integrations reducing manual processing time by 90%.
- Implemented vendor ETL automation, batch captioning and image processing pipelines, vectorization of over 2 million products enabling hybrid retrieval with autotcut.
- Deployed CLIP based embedding models (Docker, Cloudrun)
- Orchestrated handoffs, context assembly logic, wishlist management (firestore), tool and function calling, evaluation dashboard setup
- Infrastructure management and cost optimization saving upto 1100 euros per month over various deployed and subscription services.

2024 - 2025

Sagar Institute of Science Technology and Research | India

Assistant Professor

- Designed and taught graduate-level courses on Machine Learning, Deep Learning, and GenAI.
- Mentored and guided projects for over 300 students.
- Conducted research on neurocomputing - Grid and Place neuron modeling for spatial navigation.

EDUCATION

2021 - 2024 | Deggendorf Institute of Technology, Germany

M.S. in Artificial Intelligence and Data Science

2015 - 2019 | UIT-RGPV, India
Bachelor of Engineering, IT

SKILLS

Multi-agent systems, agent orchestration, tool use/function calling, planning systems, long-term memory architectures, knowledge graphs, RAG pipelines, autonomous research & evaluation loops, LLM integration (open-source & SOTA), Python, API design & integration, Docker/containerization, cloud infrastructure, vector databases, linear algebra, optimization, probability, graph theory, dynamical systems, scientific computing/simulation

ACHIEVEMENTS

Speaker | October 2026

Arabian Chatbot Conference, Abu Dhabi

Top Star Performer
Livello Technologies

2023 - 2024

Livello Technologies | Düsseldorf, Germany

Data Scientist & Researcher

- Architected a revenue and product demand forecasting system, p90 latency = 30-120 ms (single horizon, single series)
- Managed production releases on GCP, Cloud Run instances using CI/CD pipelines (Gitlab).
- Developed ETL pipelines (mongoDB, BigQuery), ensuring high quality data, integrity and accessibility for ML tasks.

2022-2023

Syskron GmbH (Krones) | Regensburg, Germany

AI Intern

- Developed self-optimizing pipelines for factory telemetry using feedback-driven learning loops, reducing drift impact by 28%.
- Co-led prototype to production transition of AI tools, managing testing, deployment, and support phases.
- Assisted factory set point setup via reinforcement learning (ContilooAI)

2018-2021

State Government of MP, India | India

Data Scientist and Analyst (Contract)

- Applied statistical analysis and SVM models to optimize resource allocation, improving system performance by 60%.
- Built chatbots using IBM Watson for policy-driven citizen services.
- Worked on structured data transformation using Python and JSON, enabling large-scale analytics.
- Built dashboards and visualizations to communicate data-driven insights to policymakers and non-technical stakeholders.

Projects

Adaptive Knowledge Engine

Reads authoritative documents into a cited, bi-temporal knowledge graph and answers three precise shapes of question against it: exact lookup, page retrieval, and configuration validity. Every answer carries a citation to its exact source page. When the engine cannot ground an answer, it stops rather than guess.

FraudHawk

Built autoencoder + Isolation Forest agent with real-time drift detection and retraining triggers.

- Evaluated feature impact using statistical methods and live KPI monitoring.

Eva: AI-Powered Health Companion

Built a real-time AI agent for evidence-based diagnostics using medical PDFs, X-rays, and clinical videos.

- Integrated LangChain + Agno Agents for retrieval-augmented generation (RAG), explainability (XAI), and on-device interaction.

MediSense-AI – Intelligent Personal Medicine Assistant

It is a personal medicine assistant that uses semantic search over symptom inputs to recommend medications with details on diseases, ingredients, dosage, and cost. Built with Streamlit and Pinecone, it offers fast, intelligent, and accessible healthcare guidance.

End-to-end revenue and product demand forecasting system

This SARIMAX based forecasting service predicts daily revenue and product demand for kiosks, aiding inventory management and resource allocation. It provides key insights like net sales, demand forecasts, revenue comparisons, and top-selling products/kiosks. The pipeline handles data migration, modeling, and forecasting

Financial Compliance Analyst

Retrieval-augmented system for spotting financial red flags from organization data.

Satellite Image Segmentation

Process satellite images quickly and efficiently to generate segmented outputs in real time.

REFERENCES

- **David Kluners – Founder and CEO** – nexwise.ai (david@nexwise.ai)
- **Markus Zöfl – Innovationsmanager AI** (M.Zoelfl@syskron.com)
- **Victor Mita – Founder** – Estyl (v.mita@estyl.ai)
- **Dr. Manish Billore – Principal** – Sagar Institute of Science and Technology (principal.sistec@sistec.ac.in)
- **Alexander Eissing – CEO** – Livello Technologies (alex@livello.com)

PUBLICATIONS

- [AI Driven Circular Supply Chain](#) (International Conference on Marketing 5.0 & Regenerative Economics 2025)
- [A comparison of time series models for revenue and product demand forecasting in smart fridges.](#)

CERTIFICATIONS

- [Product management for AI and Data Science](#): 365 Data Science
- [Certified Data Scientist](#): 365 Data Science
- [Build Chat Applications with OpenAI and LangChain](#): 365 Data Science
- [AWS Machine Learning Speciality](#): Sundog Education
- [GitLab – CI/CD Pipelines & DevOps](#): Udemy
- [Data Science Math Skills](#): Duke University
- [Time Series Analysis, Forecasting, and Machine Learning](#): Udemy
- [Vector Databases with Pinecone](#): 365 Data Science
- [GenerativeAI path](#): Google
- [Fundamentals of Docker and Kubernetes](#): Scaler